Lakshya JEE 2.0 2025

Maths

Method of Differentiation

DPP: 4

Q1 Differentiate $f(\sin x)$ w.r.t. $f(\cos x)$

(A) $-\frac{f'(\sin x)\cos x}{f'(\cos x)\sin x}$
(B) $\frac{f'(\cos x)\sin x}{f'(\sin x)\cos x}$
(C) $-\frac{f'(\cos x)\sin x}{f'(\sin x)\cos x}$
(D) $\frac{f'(\cos x)\sin x}{f'(-\sin x)\cos x}$

Q2 If $f(x)=x+\tan x$ and f is inverse of g, then $g'(x)$ equals

(A) $\frac{1}{1+[g(x)-x]^2}$
(B) $\frac{1}{2-[g(x)-x]^2}$
(C) $\frac{1}{2+[g(x)-x]^2}$
(D) None of these

Q3 Differentiate $\sin^{-1}\left(\frac{2x}{1+x^2}\right)$ w.r.t. $\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$

(A) 0
(B) 1
(C) -1
(D) $\tan x$

Q4 Let $f(x)=\exp\left(x^3+x^2+x\right)$ for any real number x and let g be the inverse function for f. The value of $g'\left(e^3\right)$ is

(A) $\frac{1}{6e^3}$
(B) $\frac{1}{6}$
(C) $\frac{1}{34e^{39}}$
(D) 6

Q5 $\frac{d^2x}{dy^2}=$

(A) $-\left(\frac{dy}{dx}\right)^{-3}\cdot\left(\frac{d^2y}{dx^2}\right)$
(B) $\frac{d^2y}{dx^2}\cdot\left(\frac{dy}{dx}\right)^{-3}$
(C) $\left(\frac{d^2y}{dx^2}\right)^2\cdot\left(\frac{dy}{dx}\right)^{-3}$
(D) $-\left(\frac{d^2y}{dx^2}\right)^2\cdot\left(\frac{dy}{dx}\right)^{-3}$

Q6 If $y=e^{2x}$, then $\left(\frac{d^2y}{dx^2}\right)\left(\frac{d^2x}{dy^2}\right)=$

(A) $\frac{2}{y}$
(B) $-\frac{2}{y}$
(C) $\frac{y}{2}$
(D) $-\frac{y}{2}$

Q7 If the function $f(x)=x^3+e^{x/2}$ and $g(x)=f^{-1}(x)$, then the value of $g'(1)$ is

Q8 If $y=\tan^{-1}\left(\frac{a\cos x-b\sin x}{b\cos x+a\sin x}\right)$, then $\frac{dy}{dx}$ is equal to

(A) 2
(B) -1
(C) $\frac{a}{b}$
(D) $\frac{b}{a}$

Q9 $\frac{d}{dx}\left[\tan^{-1}\left(\frac{a-x}{1+ax}\right)\right]$ is equal to

(A) $-\frac{1}{1+x^2}$
(B) $\frac{1}{1+a^2}-\frac{1}{1+x^2}$
(C) $\frac{1}{1+\left(\frac{a-x}{1+ax}\right)^2}$
(D) $\frac{-1}{\sqrt{1-\left(\frac{a-x}{1+ax}\right)^2}}$

Q10 If $y=x+e^x$, then $\frac{d^2x}{dy^2}$ is:

(A) e^x
(B) $-\frac{e^x}{(1+e^x)^3}$
(C) $-\frac{e^x}{(1+e^x)^2}$
(D) $\frac{-1}{(1+e^x)^3}$

Q11 If, $y=\sin x+e^x$, then $\frac{d^2x}{dy^2}=$

(A) $(-\sin x+e^x)^{-1}$
(B) $\frac{\sin x-e^x}{(\cos x+e^x)^2}$
(C) $\frac{\sin x-e^x}{(\cos x+e^x)^3}$
(D) $\frac{\sin x+e^x}{(\cos x+e^x)^3}$

Q12 $x=t\cos t, y=t+\sin t$, then $\frac{d^2x}{dy^2}$ at $t=\frac{\pi}{2}$ is:
(A) $\frac{\pi+4}{2}$
(B) $-\frac{\pi+4}{2}$
(C) -2
(D) None of these

Q13 If $x=2\sin\theta-\sin 2\theta$ and $y=2\cos\theta-\cos 2\theta, \theta\in[0,2\pi]$, then $\frac{d^2y}{dx^2}$ at $\theta=\pi$ is:
(A) $\frac{3}{8}$
(B) $\frac{3}{2}$
(C) $\frac{3}{4}$
(D) $-\frac{3}{4}$

Q14 Let $y=y(x)$ be a function of x satisfying $y\sqrt{1-x^2}=k-x\sqrt{1-y^2}$ where k is constant and $y\left(\frac{1}{2}\right)=-\frac{1}{4}$. Then $\frac{dy}{dx}$ at $x=\frac{1}{2}$, is equal to:
(A) $\frac{5}{\sqrt{7}}$
(B) $-\frac{5}{\sqrt{4}}$
(C) $-\frac{5}{\sqrt{2}}$
(D) $\frac{-\sqrt{5}}{2}$

Q15 Let $f:S\rightarrow S$ where $S=(0,\infty)$ be a twice differentiable function such that $f(x+1)=xf(x)$. If $g:S\rightarrow R$ be defined as $g(x)=\log_e f(x)$, then the value of $|g''(5)-g''(1)|$ is equal to:
(A) $\frac{205}{144}$
(B) $\frac{197}{144}$
(C) $\frac{187}{144}$
(D) 1

Q16 Let $f:R\rightarrow R, g:R\rightarrow R$ and $h:R\rightarrow R$ be differentiable functions such that $f(x)=x^3+3x+2, g(f(x))=x$, $h(g(g(x)))=x$ for all $x\in R$. Then
(A) $g'(2)=\frac{1}{15}$
(B) $h'(1)=666$
(C) $h(0)=16$
(D) $h(g(3))=36$

Answer Key

Q1	(A)
Q2	(C)
Q3	(B)
Q4	(A)
Q5	(A)
Q6	(B)
Q7	2
Q8	(B)
Q9	(A)
Q10	(B)
Q11	(C)
Q12	(B)
Q13	(A)
Q14	(D)
Q15	(A)
Q16	(B, C)